08/01/2025

Phase-2_Code-B

Aakash
Medical|IIT-JEE|Foundations

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, *Ph.*011-47623456

MM : 720 | NCERT Booster Test Series for NEET (RM)-2024-25_T03B | Time : 200 Min.

Topics Covered:
Physics: Electrostatic Potential and Capacitance, Current Electricity, Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction, Alternating Current
Chemsitry: Chemical Kinetics, Organic Chemistry: Some Basic Principles & Techniques, Hydrocarbons, Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers
Botany: Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plants
Zoology: Animal Kingdom-II: Non-chordates (Annelida to Hemichordata), Chordates, Structural Organisation in Animals– Animal Morphology (Cockroach, Frog)

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B.** You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks.** For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

EinsteinX369

PHYSICS

SECTION-A

1. Two plates are at potential –10 V and +10 V. The separation between the plates is 4 cm. The electric field between them is

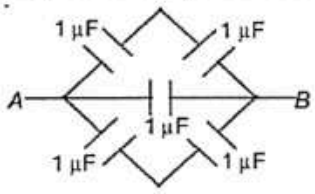
(1) 1000 N/C

(2) 600 N/C

(3) 500 N/C

(4) 300 N/C

2. The net capacitance of a system of capacitance as shown in the figure between points A and B is



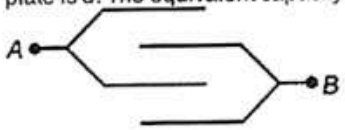
(1) 1 μF

(2) 2 μF

(3) 3 μF

(4) 4 μF

3. Four parallel plates of the same area of cross-section A are joined as shown in the figure. The distance between each plate is d. The equivalent capacity across A and B will be



(1) $\frac{2\varepsilon_0 A}{d}$

(2) $\frac{3\varepsilon_0 A}{2d}$

(3) $\frac{\varepsilon_0 A}{d}$

(4) $\frac{3\varepsilon_0 A}{d}$

4. Inside a hollow charged spherical conductor, the electric potential

(1) Is constant

(2) Varies directly as the distance from the centre

(3) Varies inversely as the distance from the centre

(4) Varies inversely as the square of the distance from the centre

5. Magnetic susceptibility is small and negative for

(1) Diamagnetic substances

(2) Paramagnetic substances

(3) Ferromagnetic substances

(4) All of these

6. A uniform magnetic field exists in the region given by $\vec{B} = \left(3\hat{i} + 4\hat{j} + 5\hat{k}\right)$T. A rod of length 1 m placed along y-axis is moved along x-axis with constant velocity 1 m/s, then induced emf in the rod is

(1) Zero

(2) 5 V

(3) 2 V

(4) 4 V

7. Energy in current carrying ideal inductor is associated with

(1) Magnetic field

(2) Electric field

(3) Gravitational field

(4) Electric and magnetic field

8. Two inductor coils of self inductance 8 mH and 18 mH are mutually inducted such that the effective flux in one coil is completely linked with the other. The mutual inductance between these coils is

(1) 10 mH

(2) 12 mH

(3) 14 mH

(4) 16 mH

9. **Assertion (A):** A pure inductor acts as perfect conductor in steady state DC circuit.
Reason (R): For steady state in DC circuit, inductor behaves as open circuit.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

10. An alternating current having peak value 10 A is used to heat a metal wire in a given time. To produce the same heating effect in same time a constant current can be used is

(1) $\frac{10}{\sqrt{2}}$ A

(2) 10 A

(3) $10\sqrt{2}$ A

(4) $\frac{5}{\sqrt{2}}$ A

11. If the charge flowing through the cross-section of a wire is given as :
$Q(t) = (2t^2 - 4t)$ C, then current in wire at $t = 2$ s is

(1) 4 A

(2) Zero

(3) 2 A

(4) 8 A

12. If current density inside a conductor is $\vec{J} = \left(2\hat{i} + \hat{k}\right)$ A/m^2 and the area through which the current flows is $\vec{A} = \left(4\hat{i} + 2\hat{j}\right)$m^2, then current through the area is

(1) 10 A

(2) 2 A

(3) 8 A

(4) 4 A

EinsteinX369

23. The work done in moving a neutron between two points having potential difference 20 V is

(1) 3.2×10^{-18} J

(2) 1.6×10^{-18} J

(3) 1.6×10^{-19} J

(4) Zero

24. Kirchhoff's loop rule is based on

(1) Law of conservation of energy

(2) Law of conservation of charge

(3) Law of conservation of momentum

(4) Wheatstone bridge

25. The magnetic flux linked with a coil varies with time (t) as $\phi(t) = (t^2 - 6t + 3)$ Wb. The induced current is zero at (time is in second)

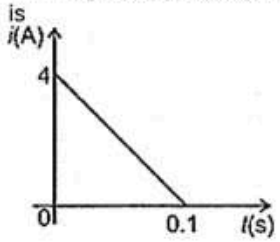
(1) $t = 6$ s

(2) $t = 3$ s

(3) $t = 1$ s

(4) $t = 2$ s

26. For a coil of resistance $10\ \Omega$, the induced current developed by changing magnetic flux through it is shown in the figure. The magnitude of change in flux through the coil (in weber) is



(1) 8

(2) 2

(3) 6

(4) 4

27. When the rate of flow of charge through a metallic conductor of non-uniform cross-section is uniform, then quantity that remains constant along the conductor is

(1) Current

(2) Current density

(3) Electric potential

(4) Electric field

28. A charged particle moves in a magnetic field $\vec{B} = 10\hat{i}$ with initial velocity $\vec{u} = 5\hat{i} + 4\hat{j}$. The path of the particle will be

(1) Straight line

(2) Circle

(3) Helical

(4) None

29. If a current of 5 A flowing in a coil of self inductance 2 mH is cut off in 10 ms, then change in magnetic flux, is

(1) 10^{-2} Wb

(2) 1 Wb

(3) 100 Wb

(4) 0.1 Wb

30. A charged particle of charge q and mass m is moving with velocity v on a circular path in uniform magnetic field B. Its time period is independent of

(1) m

(2) v

(3) q

(4) B

31. A charged particle is released from rest in a region of uniform magnetic field. The path of the particle will be

(1) Straight line

(2) Circle

(3) Helix

(4) It will not experience any force

32. An electric bulb marked 25 W and 100 V is connected across 50 V supply. Now its power is

(1) 25 W

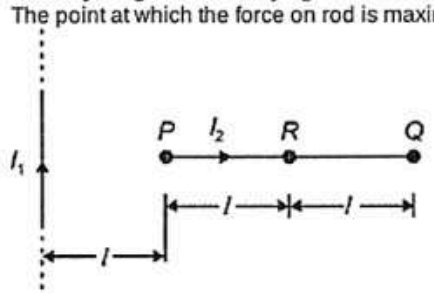
(2) 12.5 W

(3) 6.25 W

(4) 50 W

Einstein369

33. A current carrying rod PQ is placed perpendicular to an infinitely long current carrying wire as shown in the figure. The point at which the force on rod is maximum



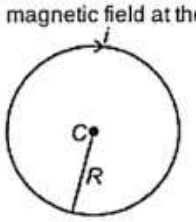
(1) P

(2) Q

(3) R

(4) Somewhere between P and R

34. Current i is flowing in a circular wire of radius R. The magnetic field at the centre is



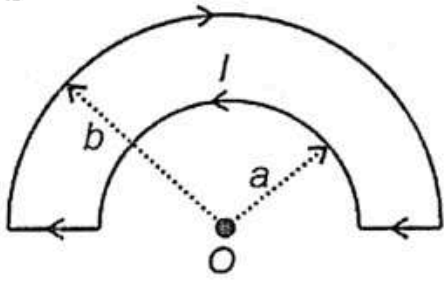
(1) $B = \frac{\mu_0 i}{2R}\otimes$

(2) $B = \frac{\mu_0 i}{4R}\otimes$

(3) Zero

(4) $B = \frac{2\mu_0 i}{R}\otimes$

35. The magnetic dipole moment of the loop as shown in figure is



(1) $\frac{\pi I(b^2-a^2)}{4}$ outward

(2) $\frac{\pi I(b^2-a^2)}{2}$ outward

(3) $\frac{\pi I(b^2-a^2)}{4}$ inward

(4) $\frac{\pi I(b^2-a^2)}{2}$ inward

EinsteinX369

SECTION-B

36. The external work required to reduce the distance between two charges $q_1 = 4\ \mu C$ and $q_2 = 20\ \mu C$ from 2 m to 1 m will be

(1) 3.6×10^{-1} J

(2) 1.8×10^{-1} J

(3) 36 J

(4) 3.8 J

37. Electrostatic potential in a region varies with coordinates x, y, z as $V = (3x - 2y + 4z)$ volt. The electric field in this region is

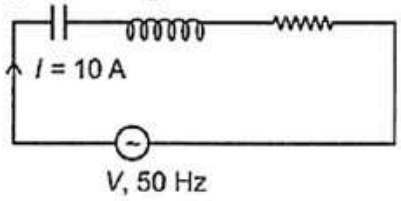
(1) Uniform

(2) Non-uniform varying with x

(3) Non-uniform varying with x and y both

(4) Non-uniform varying with x, y and z

38. Power factor of the A.C. circuit as shown in the figure is –
$V_C = 50$ V $V_L = 80$ V $R = 4\ \Omega$



(1) 0.6

(2) 0.3

(3) 0.5

(4) 0.8

39. In crossed electric and magnetic field, the velocity of charged particle which passes undeflected through the region may be (where E is electric field and B is magnetic field)

(1) $v = E^2B$

(2) $v = \frac{E}{B}$

(3) $v = \frac{E^2}{B}$

(4) $v = \frac{B}{E}$

40. The unit of intensity of magnetisation is

(1) ampere-meter

(2) ampere-(meter)^2

(3) ampere/meter

(4) ampere-(meter)^3

41. A wire of length L carries a current I, is bent in the form of circle. The magnitude of its magnetic moment is

(1) $4\pi IL^2$

(2) $\frac{2IL^2}{\pi}$

(3) $\frac{IL^2}{\pi}$

(4) $\frac{IL^2}{4\pi}$

42. Which of the following expression correctly represents the Biot-Savart's law? (symbols have their usual meanings)

(1) $\overrightarrow{dB} = \frac{\mu_0}{4\pi}\frac{I\vec{dl}\times\vec{r}}{r^3}$

(2) $\overrightarrow{dB} = \frac{\mu_0}{4\pi}\frac{I\vec{dl}\times\vec{r}}{r^2}$

(3) $\overrightarrow{dB} = \frac{\mu_0}{4\pi}\frac{I\vec{dl}\times\vec{r}}{r^3}$

(4) $\overrightarrow{dB} = \frac{\mu_0}{4\pi}\frac{I\vec{dl}\times\vec{r}}{r}$

43. The susceptibility of certain magnetic material is 400. What is the class of magnetic material?

(1) Diamagnetic

(2) Paramagnetic

(3) Ferromagnetic

(4) Fero electric

44. The resistance of galvanometer is 20 Ω and the maximum current which can be passed through it is 0.004 A. The shunt resistance that should be connected to it to convert it into an ammeter of range 0–0.1 A nearly is

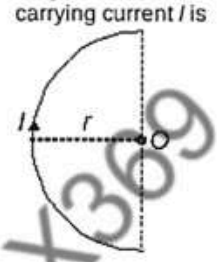
(1) 0.2 Ω

(2) 0.02 Ω

(3) 0.08 Ω

(4) 0.8 Ω

45. Magnetic field at the centre O of semicircular arc of radius r carrying current I is



(1) Zero

(2) $\frac{\mu_0 I}{4r}$

(3) $\frac{\mu_0 I}{2r}$

(4) $\frac{\mu_0 I}{8r}$

46. In the AC circuit, the current is expressed as $I = 100\sin(200\pi t)$. In this circuit, the current rises from zero to half the peak value in time (where t is in s and I is in A)

(1) $\frac{1}{400}$s

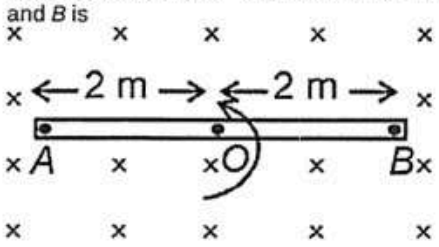
(2) $\frac{1}{1200}$s

(3) $\frac{1}{800}$s

(4) $\frac{1}{1600}$s

EinsteinX369

47. The given rod *AB* is rotated with an angular speed of 20 rpm in a uniform transverse magnetic field of magnitude 10 T, about its centre O. The emf induced between points *A* and *B* is



(1) $\frac{40\pi}{3}$V

(2) $\frac{160\pi}{3}$V

(3) $\frac{80\pi}{3}$V

(4) Zero

48. Ohm-metre is the unit of

(1) Resistance

(2) Conductance

(3) Resistivity

(4) Potential

49. The rate of increase of current through an inductor is 2.4 A/s. If voltage drop across the inductor is 1.6 V then self inductance of the inductor will be

(1) $\frac{2}{3}$H

(2) $\frac{1}{3}$H

(3) $\frac{3}{2}$H

(4) $\frac{1}{2}H$

50. Wattless elements are

(1) Pure inductor

(2) Pure capacitor

(3) Pure resistor

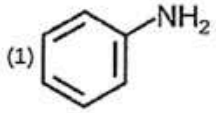
(4) Both (1) and (2)

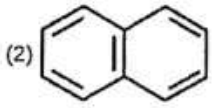
EinsteinX369

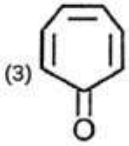
CHEMISTRY

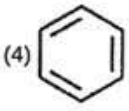
SECTION-A

51. Which among the following is a non-benzenoid compound?

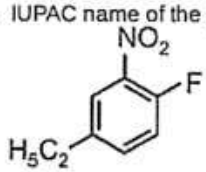
(1) 

(2) 

(3) 

(4) 

52. IUPAC name of the given compound is



(1) 1-Ethyl-4-fluoro-3-nitrobenzene

(2) 5-Ethyl-2-fluoro-1-nitrobenzene

(3) 4-Ethyl-1-fluoro-2-nitrobenzene

(4) 3-Ethyl-6-fluoro-1-nitrobenzene

53. 0.32 g of sulphur containing organic compound gave 0.64g of barium sulphate in sulphur estimation. The percentage of sulphur in the compound is [Molar mass of $BaSO_4$ = 233 g/mol]

(1) 42.2%

(2) 27.5%

(3) 28.4%

(4) 32.5%

54. $CH_3CH_2CH_2OCH_3$ and $CH_3CH_2OCH_2CH_3$ are

(1) Metamers

(2) Tautomers

(3) Chain isomers

(4) Functional isomers

55. Compound having most acidic hydrogen is

(1) 

(2) 

(3) 

(4) 

56. Which among the following is a temporary effect?

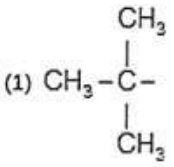
(1) Inductive effect

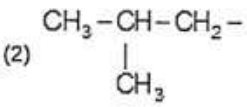
(2) Electromeric effect

(3) Resonance

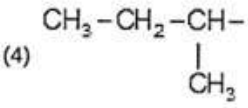
(4) Hyperconjugation

57. The structure of sec-butyl group is

(1) 

(2) 

(3) $CH_3CH_2CH_2CH_2-$

(4) 

58. The group which shows –R effect is

(1) $-NO_2$

(2) –Cl

(3) –OH

(4) $-CH_3$

59. In the detection of P, yellow ppt obtained is due to formation of

(1) $(NH_4)_3PO_4$

(2) $(NH_4)_3PO_4 \cdot MoO_4$

(3) $(NH_4)_3PO_4 \cdot 12MoO_3$

(4) $12(NH_4)_3PO_4 \cdot MoO_3$

60. CaC_2 on hydrolysis gives

(1) CH_4

(2) C_2H_6

(3) C_2H_2

(4) C_2H_4

61. During sulphonation of Benzene, the electrophile is

(1) SO_2

(2) H_2SO_3

(3) H_3O^+

(4) SO_3

62. Which compound will not decolourise Br_2/H_2O?

(1) 

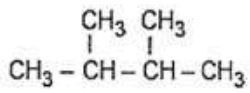
(2) 

(3) 

(4) $CH_3 - C \equiv CH$

63. n-Hexane on heating to 773 K at 10–20 atm pressure in presence of vanadium pentaoxide forms

(1) $CH_2 = CH - CH_2 - CH_2 - CH_2 - CH_3$

(2) 

(3) 

(4) 

EinsteinX369

64. Major product formed in the given reaction is

$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$

(1) CH_3OH

(2) $CH_3–O–CH_3$

(3) HCOOH

(4) HCHO

65. Consider the reaction,

$H_3C-\underset{\underset{CH_3}{|}}{C}=CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$ Products

These products are

(1) Propan-2-one, methanoic acid

(2) Propanoic acid, Methanal

(3) Propan-2-one, Methanal

(4) Propanoic acid, Methanoic acid

66. $CH_3-CH=CH_2 \xrightarrow[(ii)\ H_2O_2/OH^-]{(i)\ B_2H_6\text{-}THF} A$

Product A is

(1) CH_3-CH_2-CHO

(2) $CH_3-CH_2-CH_2-OH$

(3) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

(4) $CH_3-\underset{\underset{O}{||}}{C}-CH_3$

67. Maximum boiling point among the following is of

(1) n-butane

(2) n-pentane

(3) Isopentane

(4) Neopentane

68. Time required for 100% completion of a zero order reaction is ($[A]_0$ is the initial concentration of reactant)

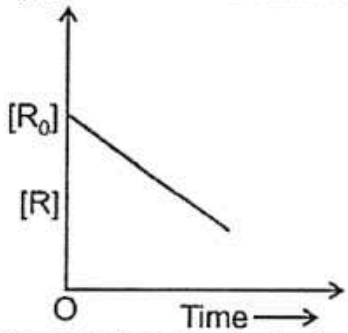
(1) $\frac{[A]_0}{k}$

(2) $k[A]_0$

(3) $\frac{k}{[A]_0}$

(4) $2k[A]_0$

69. A graph for a zero order reaction is represented as



The slope of graph is given by (k = rate constant)

(1) $\frac{1}{k}$

(2) $\frac{1}{[R]_0}$

(3) –k

(4) $\frac{-1}{k}$

70. In a zero order reaction for every 10° rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 50°C, the rate of the reaction will become

(1) 64 times

(2) 32 times

(3) 8 times

(4) 16 times

71. **Incorrect** statement about molecularity of a reaction is

(1) Molecularity cannot be zero.

(2) Molecularity is applicable only for elementary reaction.

(3) For complex reaction, molecularity has no meaning.

(4) Molecularity can be a non-integer

72. For a reaction 2A + B → 3C, the rate of appearance of C at time t is 1.2×10^{-4} mol $L^{-1}s^{-1}$. The rate of reaction is

(1) 2×10^{-5} mol $L^{-1}s^{-1}$

(2) 4×10^{-5} mol $L^{-1}s^{-1}$

(3) 4×10^{-6} mol $L^{-1}s^{-1}$

(4) 2×10^{-6} mol $L^{-1}s^{-1}$

73. For a chemical reaction, R → P

if $\Delta H = -75$ kJ mol^{-1} and activation energy for the forward reaction is 50 kJ mol^{-1} then activation energy for the backward reaction will be

(1) 125 kJ mol^{-1}

(2) 25 kJ mol^{-1}

(3) 100 kJ mol^{-1}

(4) 150 kJ mol^{-1}

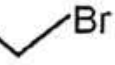
EinsteinX360

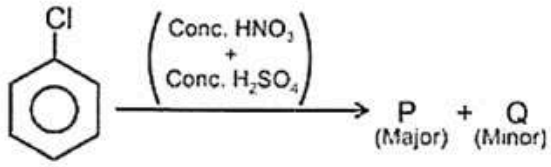
74. Which among the following will react fastest by S_N1 mechanism?

(1) 

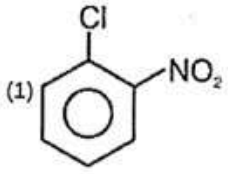
(2) 

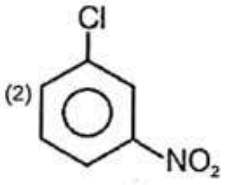
(3) 

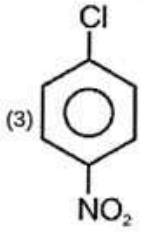
(4) 

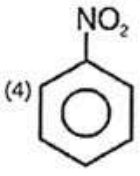
75. 

The major product (P) is

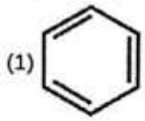
(1) 

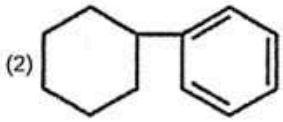
(2) 

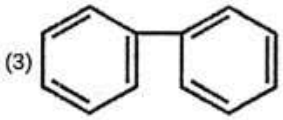
(3) 

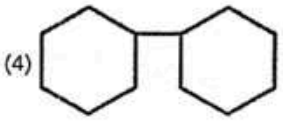
(4) 

76. When chlorobenzene is treated with sodium in presence of dry ether it gives

(1) 

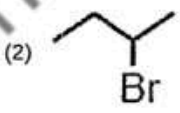
(2) 

(3) 

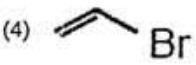
(4) 

77. Nucleophilic substitution reaction is most difficult in

(1) 

(2) 

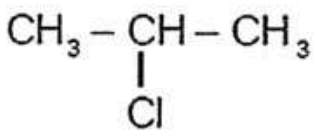
(3) 

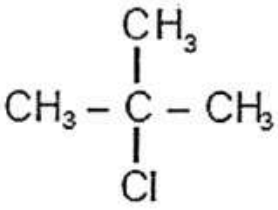
(4) 

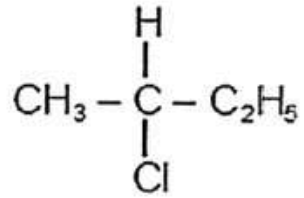
EinsteinX369

78. The compound which will exist as a pair of enantiomer is

(1) $CH_3 - CH_2 - Cl$

(2) 

(3) 

(4) 

79. Swart's reaction is used to prepare

(1) Alkyl fluorides

(2) Alkyl chlorides

(3) Alkyl bromides

(4) Alkyl iodides

80. The compound A on treatment with Na gives B, and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order

(1) $C_2H_5OH, C_2H_6, C_2H_5Cl$

(2) $C_2H_5OH, C_2H_5Cl, C_2H_5ONa$

(3) $C_2H_5OH, C_2H_5ONa, C_2H_5Cl$

(4) $C_2H_5Cl, C_2H_6, C_2H_5OH$

81. The electrophile involved in Reimer-Tiemann reaction is

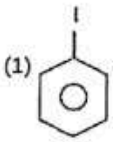
(1) $\overset{\oplus}{C}HCl_2$

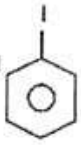
(2) $:CCl_2$

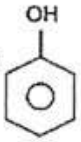
(3) $\overset{\ominus}{C}Cl_3$

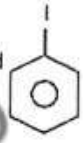
(4) $\overset{\oplus}{C}HO$

82. When anisole is treated with one equivalent HI, the products formed are

(1)  and CH_3–OH

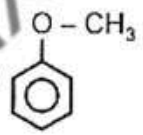
(2) $CH_3 - I$ and 

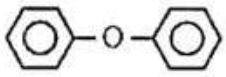
(3) $CH_3 - I$ and 

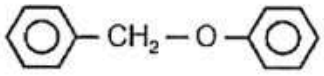
(4) $C_2H_5 - OH$ and 

83. Which of the following ethers cannot be prepared by Williamson's synthesis?

(1) CH_3–O–CH_3

(2) 

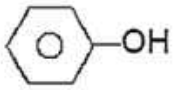
(3) 

(4) 

EinsteinX269

84. The reaction of Lucas reagent is fastest with

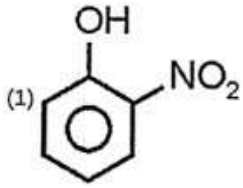
(1) CH_3CH_2-OH

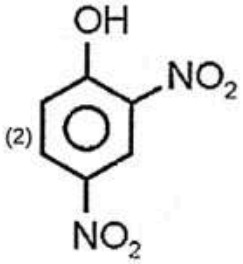
(2) 

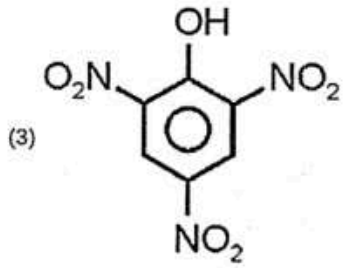
(3) $(CH_3)_2CH-OH$

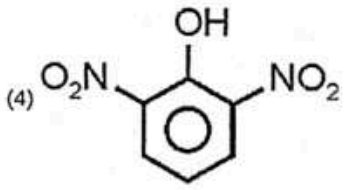
(4) $(CH_3)_3C-OH$

85. Picric acid is

(1) 

(2) 

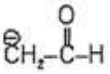
(3) 

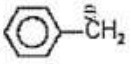
(4) 

EinsteinX369

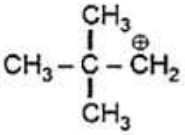
SECTION-B

86. Which of the following species is not stabilised by resonance?

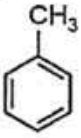
(1) 

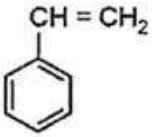
(2) 

(3) $CH_2 = CH - \overset{\ominus}{C}H_2$

(4) 

87. In which of the following molecules, all atoms are coplanar?

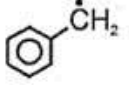
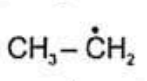
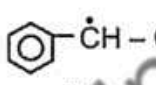
(1) 

(2) 

(3) $CH_3 - CH = CH_2$

(4) $CH_2 = C = CH_2$

88. Arrange the following free radicals in the order of stability.

I II III IV

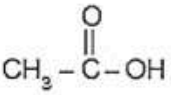
(1) I > II > III > IV

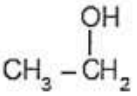
(2) IV > I > III > II

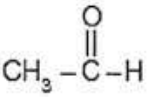
(3) IV > I > II > III

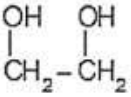
(4) III > IV > II > I

89. Ethene on reaction with Baeyer's reagent forms

(1) 

(2) 

(3) 

(4) 

90. Among the following molecules which gives only one monochloro derivative?

(1) n-Butane

(2) Iso-butane

(3) Neopentane

(4) Propane

91. Which among the following alkanes can be made in good yield by Wurtz reaction?

(1) n-pentane

(2) n-propane

(3) n-heptane

(4) n-hexane

92. When the temperature of reaction is increased from 27°C to 37°C, the rate constant becomes doubled. The energy of activation for the reaction is

(1) 300 R ln 2

(2) 9300 R ln 2

(3) 310 R ln 3

(4) 410 R ln 2

93. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**:

Assertion (A) : For a first order reaction $t_{99.9\%} = 10 \times t_{50\%}$.

Reason (R) : Half life of first order reaction is independent of initial concentration of reactants.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

94. For a first order reaction A $\rightarrow$B, the ratio of time required for 99.9% completion to the time required for 99.99% completion is

(1) $\frac{3}{4}$

(2) $\frac{1}{5}$

(3) $\frac{1}{10}$

(4) $\frac{2}{3}$

95. 2-Bromo-3-methylpentane is heated with potassium ethoxide in ethanol. The major product obtained is

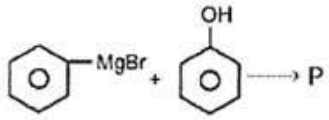
(1) 3-Methylpent-2-ene

(2) 2-Ethoxy-3-methylpentane

(3) 2-Methylpent-2-ene

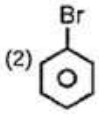
(4) 4-Methylpent-2-ene

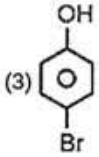
Einstein 869

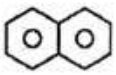
96. 

Product (P) is

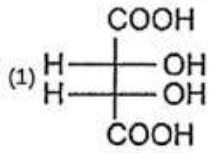
(1) 

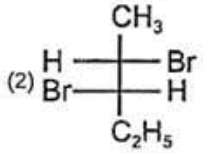
(2) 

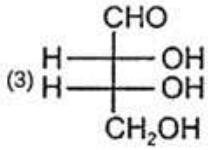
(3) 

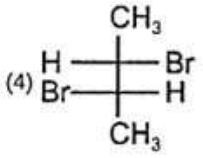
(4) 

97. Which among the following is a meso compound?

(1) 

(2) 

(3) 

(4) 

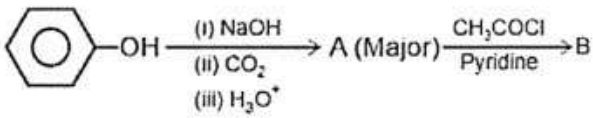
98. Which alcohol in reaction with Cu at 573 K gives ketone as major product?

(1) 

(2) 

(3) 

(4) 

99. 

Here, A and B respectively are

(1) Salicylic acid, aspirin

(2) Salicylic acid, salol

(3) Salicylic acid, oil of winter green

(4) Salicylaldehyde, aspirin

100. Given below are two statements: One is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A): Boiling point of ethanol is higher than methoxymethane and propane.
Reason (R): Intermolecular hydrogen bonding is present in alcohols which lacks in ethers and hydrocarbons.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

BOTANY

EinsteinX369

SECTION-A

101. The following diagram represents L.S. of an embryo of grass



The labelled part A is

(1) Coleoptile

(2) Scutellum

(3) Epiblast

(4) Radicle

102. The pigment that appear blue-green in the chromatogram is

(1) Chlorophyll *a*

(2) Chlorophyll *b*

(3) Carotene

(4) Xanthophyll

103. Read the given Assertion and Reason, and choose the **correct** option.

Assertion (A): The splitting of the water molecule takes place on the inner side of thylakoid membrane.

Reason (R): Protons or hydrogen ions that are produced by the splitting of water accumulate within the lumen of the thylakoids.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion

(3) Only Assertion is true

(4) Both Assertion and Reason are false

104. Cyclic photophosphorylation

(1) Requires an external electron donor

(2) Is performed by collaboration of both PS-I and II

(3) Synthesizes ATP only

(4) Fixes CO_2 in green plants only

105. All of the given factors are plant factors which affect photosynthesis, **except**

(1) Age of leaves

(2) Orientation of leaves

(3) Availability of water

(4) Amount of chlorophyll

106. First stable product of C_3 cycle is

(1) 3-PGA

(2) OAA

(3) Pyruvic acid

(4) Glyceraldehyde-3-Phosphate

107. First action spectrum of photosynthesis was described through the experiments conducted by

(1) Julius von Sachs

(2) Jan Ingenhousz

(3) C.V. Niel

(4) T.W. Engelmann

108. Chemiosmosis does **not** require

(1) Electron gradient

(2) Membrane

(3) Proton pump

(4) Proton gradient

109. RuBisCO enzyme is present in all, **except**

(1) Mesophyll cells of C_3 plants

(2) Mesophyll cells of C_4 plants

(3) Bundle sheath cells of C_4 plants

(4) In CAM plants

Einstein7369

110. Identify the **incorrect** statement w.r.t. E.M.P pathway.

(1) It occurs in the cytoplasm of all eukaryotes and prokaryotes

(2) ATP is utilised in the step, which require hexokinase enzyme

(3) The conversion of 1,3-bisphosphoglycerate to 3-phosphoglyceric acid is an energy yielding process

(4) Two molecules of ATP is utilised during conversion of PEP to pyruvic acid

111. Which of the following is the favoured substrate for respiration?

(1) Organic acids

(2) Glucose

(3) Fats

(4) Proteins

112. In oxidative phosphorylation, complex V is

(1) ATP synthase

(2) Succinate dehydrogenase

(3) Cytochrome bc_1

(4) NADH dehydrogenase

113. How many complexes participate in transfer of electrons via electron transport system in mitochondria?

(1) Three

(2) Four

(3) Seven

(4) Six

114. In anaerobic organisms, glycolysis occurs in

(1) Mitochondrial matrix

(2) Cytoplasm

(3) Mesosome

(4) Mitochondrial membrane

115. How many redox equivalents are removed during conversion of one molecule of 3-PGAL into 1,3 bisphosphoglyceric acid?

(1) One

(2) Two

(3) Three

(4) Four

116. The F_1 head piece in mitochondria is a

(1) Catalytic site which produces ADP from ATP

(2) Channel through which protons cross the outer membrane

(3) Integral membrane protein complex

(4) Peripheral membrane protein complex

117. Apical dominance can be overcome by the application of

(1) Ethylene

(2) Kinetin

(3) Gibberellins

(4) Auxins

118. Maturation phase of growth is characterised by

(1) Cells show protoplasmic modifications

(2) Cell elongation and vacuolation

(3) Large conspicuous nucleus and vacuolated cytoplasm

(4) Presence of actively dividing cells

119. Read the following statements and state **True** (T) or **False** (F).

A. Relative growth rate is the measurement and comparison of total growth per unit time.

B. Primary growth is characteristic of all types of plants.

C. Auxin, ethylene and ABA in general are the plant growth inhibitors.

D. Cytokinins delays cell growth while auxin is involved only in cell division.

	A	B	C	D
(1)	T	T	F	F
(2)	F	T	F	F
(3)	F	F	T	F
(4)	F	T	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

120. Hastening the maturity period in conifers and thus, leading to early seed production can be observed by spraying

(1) Abscisic acid

(2) Ethylene

(3) Auxins

(4) Gibberellins

121. Growth of zygote in early stage

(1) Is arithmetic growth

(2) Shows growth rate where every cell divides

(3) Is explained by $L_t = L_0 + rt$ equation

(4) Leads to formation of a linear curve

122. Which hormone helps to prevent fruit and leaf drop at early stages but promotes abscission of older or mature leaves?

(1) Gibberellins

(2) ABA

(3) Auxin

(4) Ethylene

Einstein Prog

123. Efficiency index is the measure of

(1) Ability of the plant to produce new plant material

(2) Time of growth

(3) Number of cells that lose capacity to divide

(4) Rate of differentiation of plant cells

124. Select the **wrong** statement w.r.t ethylene

(1) Inhibits senescence and abscission

(2) Breaks seed and bud dormancy

(3) Indúces flowering in mango

(4) Accelerates abscission in flowers and fruits like cotton, cherry and walnut

125. ABA is also called stress hormone because

(1) It increases the stress conditions for the plants

(2) It causes the opening of stomata

(3) It increases the tolerance of plants in adverse environmental conditions

(4) It increases the water absorption capacity of the plant

126. Egg apparatus is

(1) 7 celled 8 nucleated

(2) 8 celled 7 nucleated

(3) 4 celled 3 nucleated

(4) 3 celled 3 nucleated

127. Function of filiform apparatus is to :

(1) Recognize the suitable pollen at stigma

(2) Stimulate division of generative cell

(3) Produce nectar

(4) Guide the entry of pollen tube

128. Pollen grain of angiosperms represents

(1) Male gamete

(2) Microsporangium

(3) Male gametophyte

(4) Megaspore

129. Gynoecium of *Michelia* is

(1) Monocarpellary syncarpous

(2) Multicarpellary syncarpous

(3) Monocarpellary apocarpous

(4) Multicarpellary apocarpous

130. In water lily, pollination is **not** carried out by water because

(1) Their flowers are large sized

(2) Their flowers are packed into inflorescence

(3) Their flowers emerge above the level of water

(4) Nectaries are absent in their flowers

131. How apomixis can be beneficial to the hybrid seed industry?

(1) Seeds will become too expensive for the farmers

(2) Progeny plants will not be able to maintain hybrid characters

(3) There will be no segregation of characters in the hybrid progeny

(4) Hybrid seed cultivation will decrease productivity

132. Read the following statements and select the **correct** option.

Statement (a) : Vegetative cell of pollen grain is bigger in size and rich in food reserve.

Statement (b) : Generative cell is small and contain many nucleus.

(1) Only (a) is correct

(2) Only (b) is correct

(3) Both (a) and (b) are correct

(4) Both (a) and (b) are incorrect

133. Identify the given statements as true **(T)** or false **(F)** and choose the option accordingly.

A. In an idealised sigmoid growth curve for cell in culture, stationary phase cannot be observed.

B. Increase in dry weight can also be a parameter to measure growth.

C. Development is the sum of two processes *i.e.*, growth and differentiation.

	A	B	C
(1)	T	F	T
(2)	F	F	T
(3)	F	T	T
(4)	T	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

Einstein

134. Which of the following types of pollination is functionally cross pollination and genetically similar to self pollination?

(1) Xenogamy

(2) Autogamy

(3) Geitonogamy

(4) Cleistogamy

135. Read the following statements and select the **correct** option.
Assertion (A): Maize is a wind-pollinated plant.
Reason (R): Light and non-sticky pollen grains surrounded by mucilaginous covering, large feathery stigma and single ovule in each ovary are found in wind-pollinated flowers.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

EinsteinX369

SECTION-B

136. Hilum represents the junction between

(1) Funicle and ovary wall

(2) Ovule and funicle

(3) Chalaza and integument

(4) Funicle and micropyle

137. PEN is the result of

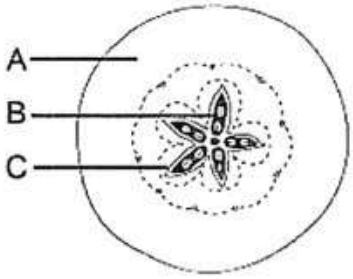
(1) Apomixis

(2) Triple fusion

(3) Polyembryony

(4) Syngamy

138. Identify the given figure with **correct** labelling :



(1) True fruit of mango, A – Mesocarp

(2) True fruit of apple, B – Seed

(3) False fruit of strawberry, C – Epicarp

(4) False fruit of apple, B – Seed

139. Parthenocarpy and apomixis are similar as both lack

(1) Fertilization process

(2) Embryo formation

(3) Fruit formation

(4) Development of seeds

140. Which one is a **wrong** statement about 'outbreeding devices'?

(1) Release of pollen and stigma receptivity are not synchronised

(2) Anther and stigma are placed at same positions

(3) Presence of unisexual flowers

(4) Presence of physical barrier between the both sexes

141. The path of carbon dioxide in the photosynthetic dark reaction was successfully traced by using

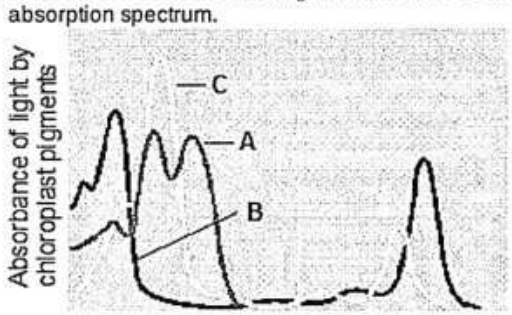
(1) Trace elements

(2) O^{18} isotope of oxygen

(3) C^{14} isotope of carbon

(4) Non-radioactive N^{15}

142. Choose the **correct** labelling for A, B and C showing the absorption spectrum.



(1) **A**-Chlorophyll-*b*, **B**-Chlorophyll-*c*, **C**-Carotenoids

(2) **A**-Carotenoids, **B**-Chlorophyll-*a*, **C**-Chlorophyll-*b*

(3) **A**-Chlorophyll-*a*, **B**-Chlorophyll-*b*, **C**-Carotenoids

(4) **A**-Carotenoids, **B**-Chlorophyll-*b*, **C**-Chlorophyll-*a*

143. 'Shield pigments' during photosynthesis

(1) Produce only ATP directly by absorbing light

(2) Protect the plant from excessive heat and also function as primary pigments

(3) Prevent photo-oxidation of chlorophyll *a*

(4) Produce both ATP and NADPH directly by absorbing light

144. The first irreversible reaction of EMP pathway is catalyzed by

(1) Phosphofructokinase

(2) Pyruvate kinase

(3) Hexokinase

(4) Aldolase

145. The terminal electron acceptor of mitochondrial ETS is

(1) H_2O

(2) O_2

(3) ATPase

(4) Cytochrome c oxidase

146. Which of the following statements is **correct** w.r.t cytochrome *c*?

(1) It is a mobile carrier for transfer of electrons between complex II and III

(2) It is a mobile carrier between complex I and IV

(3) It is a small protein attached to the outer surface of the inner mitochondrial membrane

(4) It serves the role of complex I

147. Plant, which shows different shapes of leaves according to its habitat, is

(1) Cotton

(2) Larkspur

(3) Buttercup

(4) Coriander

148. Auxin

(1) Term was derived from Latin word '*auxein*'

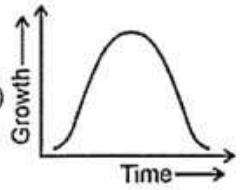
(2) Is generally produced by mature parts of plant

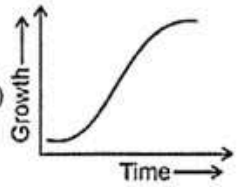
(3) Was first isolated from human urine

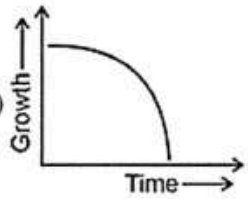
(4) Has been rarely used in agriculture and horticulture practices

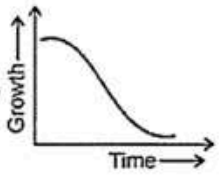
Einstein369

149. The growth curve which is characteristic of living organisms growing in a natural environment can be represented as

(1) 

(2) 

(3) 

(4) 

150. Natural cytokinin is synthesized in all the following regions of a flowering plant, **except**

(1) Root apices

(2) Developing shoot buds

(3) Old senescent leaves

(4) Young fruits

EinsteinX369

ZOOLOGY

SECTION-A

151. The largest and the second largest phylum of animal kingdom respectively are

(1) Chordata and Arthropoda

(2) Porifera and Mollusca

(3) Arthropoda and Mollusca

(4) Annelida and Arthropoda

152. Common feature between *Saccoglossus* and *Anopheles* are all, **except**

(1) Open circulatory system

(2) Complete digestive tract

(3) Multicellularity

(4) Body divided into head, thorax and abdomen

153. Choose the **correct** option to complete the analogy.
Echinoderm : *Asterias* :: Arthropod : _____.

(1) *Meandrina*

(2) *Laccifer*

(3) *Pila*

(4) *Antedon*

154. Which group of animals belong to the same phylum?

(1) *Pila*, *Pinctada*, *Saccoglossus*

(2) *Echinus*, *Cucumaria*, *Asterias*

(3) *Ophiura*, *Octopus*, *Dentalium*

(4) *Echinus*, *Locusta*, *Pheretima*

155. Coelom in *Balanoglossus* is lined by

(1) Ectoderm

(2) Mesoderm

(3) Endoderm

(4) Ectomesoderm

156. Common feature among *Nereis*, *Culex*, *Sepia* and *Ascaris* is

(1) Presence of closed circulatory system

(2) Presence of calcareous shell

(3) Absence of bilateral symmetry

(4) Presence of triploblastic condition

157. A common characteristic of earthworm, prawn and cockroach is

(1) Prominent cutaneous respiration

(2) Metameric segmentation

(3) Chitinous exoskeleton

(4) Open circulatory system

158. Which among the following statements is **not** applicable for sea cucumber?

(1) Pharyngeal gill slits are absent

(2) Notochord is absent

(3) Post-anal tail is present

(4) Presence of organ-system level of organization

159. In which one of the following the genus name, its two characters and its phylum are **not** correctly matched?

	Genus name	Two characters	Phylum
(1)	*Ophiura*	(i) Calcareous endoskeleton (ii) Water vascular system	Echinodermata
(2)	*Culex*	(i) Chitinous exoskeleton (ii) Compound eye	Arthropoda
(3)	*Nereis*	(i) Possess parapodia (ii) Monoecious	Annelida
(4)	*Pila*	(i) Soft body (ii) Respiration and excretion by gills	Mollusca

(1) (1)

(2) (2)

(3) (3)

(4) (4)

160. Read the following statements A and B and choose the **correct** answer

Statement-A : Members of the phylum Arthropoda are similar to that of Mollusca as both have ventral, solid nerve cord.

Statement-B : Radula is file-like rasping organ for feeding found in molluscs and echinoderms.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

161. The common character between *Locusta* and *Chaetopleura* is presence of

(1) Muscular foot

(2) Jointed appendages

(3) Segmentation

(4) True coelom

162. In birds, which of the following structures has/have scales?

(1) Forelimbs

(2) Skin

(3) Hindlimbs

(4) Feathers

163. Which of the following animals is characterised by the presence of gizzard in its digestive system?

(1) *Pteropus*

(2) *Neophron*

(3) *Balaenoptera*

(4) *Delphinus*

Einstein 869

164. Birds can be differentiated from mammals on the basis of presence of

(1) Four chambered heart

(2) Hollow bony endoskeleton with air cavities

(3) Homeothermy

(4) Pulmonary respiration

165. How many organisms given in the box below are oviparous?

Aptenodytes, Ornithorhynchus, Macropus, Delphinus, Balaenoptera

Select the correct option.

(1) 4

(2) 3

(3) 2

(4) 1

166. Match the items in column-I with those in column-II and choose the option with only **correct** matches.

	Column-I		**Column-II**
A.	*Torpedo*	(p)	Viviparous, fold of skin in between digits
B.	*Trygon*	(q)	Limbless, tailed amphibian
C.	*Ichthyophis*	(r)	Presence of placoid scales and electric organ
D.	*Pteropus*	(s)	Presence of poisonous sting and ventral mouth

(1) A(p), B(q), C(r), D(s)

(2) A(r), B(s), C(p), D(q)

(3) A(r), B(s), C(q), D(p)

(4) A(s), B(r), C(p), D(q)

167. Read the following statements **A** and **B** and choose the **correct** answer.
Statement A : All vertebrates are chordates but all chordates are not vertebrates.
Statement B : All mammals are homeothermous but all homeothermous animals are not mammals.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

168. In all of the following categories of organisms, syngamy generally occurs inside their body, **except**

(1) Cartilaginous fishes

(2) Reptiles

(3) Birds

(4) Bony fishes

169. How many animals given in the box below are poikilothermic?

Salamandra, Crocodilus, Pavo, Ornithorhynchus, Macropus, Labeo

Select the **correct** option.

(1) One

(2) Two

(3) Three

(4) Four

170. Read the following statements and select the **correct** option.
Statement A: Male sharks bear a pair of claspers attached to the pectoral fins that help in copulation.
Statement B: Sea horse is viviparous in nature.

(1) Both statements are incorrect

(2) Statement (A) is correct and (B) is incorrect

(3) Statement (B) is correct and (A) is incorrect

(4) Both statements are correct

171. Cranium and vertebral column are cartilaginous in organisms of class

(1) Amphibia

(2) Reptilia

(3) Aves

(4) Cyclostomata

172. All the given options are **incorrect** about *Branchiostoma*, **except**

(1) Notochord extends from head to tail region

(2) Notochord is present only in larval tail

(3) Notochord do not persist throughout their life

(4) Gill slits are absent

173. Division Agnatha is differentiated from Gnathostomata based on the absence of

(1) Fins

(2) Jaws

(3) Notochord

(4) Closed circulation

Einstein 369

174. Select the **mismatch**.

(1) *Pterophyllum* – Angelfish

(2) *Bangarus* – Krait

(3) *Macropus* – Flying fox

(4) *Ichthyophis* – Limbless amphibian

(1) (1)

(2) (2)

(3) (3)

(4) (4)

175. *Chelone* belongs to the class Reptilia. What does the class name refer to?

(1) Dual life of this vertebrate

(2) Creeping mode of locomotion

(3) Presence of cornified skin

(4) Ability to shed scales as skin cast

176. Frog eyes are protected under water by the presence of

(1) Retina

(2) Cornea

(3) Nictitating membrane

(4) Lens

177. In male cockroach, mushroom-shaped gland is present in which of the following abdominal segments?

(1) $3^{rd} - 4^{th}$

(2) $4^{th} - 5^{th}$

(3) $5^{th} - 6^{th}$

(4) $6^{th} - 7^{th}$

178. Read the following statements and choose the **correct** option.
Statement A : In cockroach, vision is known as mosaic vision with less sensitivity but more resolution
Statement B : In male cockroach, a pair of spermatheca is present in the 6th segment which opens into the genital chamber

(1) Both statements are incorrect

(2) Both statements are correct

(3) Statement A is correct but B is incorrect

(4) Statement B is correct but A is incorrect

179. The correct sequence of life cycle of frog is

(1) Fertilised egg → Adult → Tadpole

(2) Tadpole → Egg → Adult

(3) Fertilised egg → Tadpole → Adult

(4) Adult → Tadpole → Egg

180. Identify the special venous connection shared by both frogs and humans which is present between liver and intestine. Select the **correct** option w.r.t it.

(1) Renal portal system

(2) Hepatic portal system

(3) Hypophyseal portal system

(4) Lymphatic system

181. The respiratory system of cockroach consists of a network of __A__, that open through __B__ of small holes, present on the __C__ side of the body.
Select the option which fills these blanks **correctly**.

	A	B	C
(1)	Trachea	10 pairs	Ventral
(2)	Gills	8 pairs	Lateral
(3)	Trachea	8 pairs	Dorsal
(4)	Trachea	10 pairs	Lateral

(1) (1)

(2) (2)

(3) (3)

(4) (4)

182. Read the following statements w.r.t frog and choose the **correct** option.
Statement A : Testes is attached to the upper part of kidney by a double fold of peritoneum, called mesorchium.
Statement B : The vasa efferentia run transversely through the mesorchium, enter the kidneys and open into the Bidder's canal.

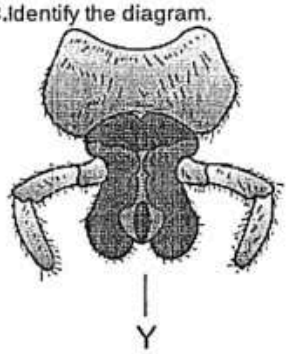
(1) Both statements A and B are incorrect

(2) Both statements A and B are correct

(3) Statement A is correct but B is incorrect

(4) Statement A is incorrect but B is correct.

183. Identify the diagram.



Which of the following mouth part of cockroach does 'Y' depict?

(1) Labrum

(2) Labium

(3) Maxilla

(4) Hypopharynx

Einstein369

184. Nocturnal, omnivores, resident of human homes, serious pests, vectors of several diseases, live in damp place, chitinous exoskeleton, piercing and sucking type mouth parts.
How many of the above mentioned characters are applicable to *Periplaneta americana*?

(1) Eight

(2) Seven

(3) Four

(4) Two

185. Comprehend the given statements.
Statement (A) : In some countries, the muscular legs of frogs are used as food by man.
Statement (B) : Frogs maintain ecological balance because they serve as an important link of food chain and food web in the ecosystem.
Select the **correct** option.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct.

(4) Only statement (B) is correct

EinsteinX369

SECTION-B

186. The frog's body is divisible into

(1) Head, Neck, Abdomen

(2) Head, Neck, trunk

(3) Head, trunk

(4) Head, Neck

187. How many cranial nerves arise from the brain in *Rana tigrina*?

(1) 8 pairs

(2) 10 pairs

(3) 12 pairs

(4) 31 pairs

188. **Assertion :** Visceral organs located in the haemocoel are bathed in haemolymph in *Periplaneta*.
Reason : Cockroach has an open circulatory system and blood in haemocoel transports the oxygen to different parts of body.
In the light of above statements, choose the **correct** answer.

(1) Both Assertion and Reason are true and reason is the correct explanation of the assertion

(2) Both Assertion and Reason are true but reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

189. Sexual dimorphism during breeding season is distinct in *Rana tigrina* due to the presence of

(1) Copulatory pad in male

(2) Nuptial pad in female

(3) Vocal cord in male

(4) Vocal sac in both male and female

190. In male frogs, a small, median chamber that is used to pass faecal matter, urine and sperms to the exterior is

(1) Fat body

(2) Cloaca

(3) Urinary bladder

(4) Ureter

191. Mouth of *Petromyzon* and *Scoliodon* is respectively

(1) Terminal and suctorial

(2) Suctorial and terminal

(3) Jawless and terminal

(4) Jawless and ventral

192. Skin of *Corvus* is dry without glands except the __A__ present at the __B__. Choose the option that fills the blanks **correctly**.

	A	B
(1)	Sweat gland	Base of the tail
(2)	Sebaceous gland	Mouth region
(3)	Oil gland	Base of the tail
(4)	Oil gland	Beak

(1) (1)

(2) (2)

(3) (3)

(4) (4)

193. All of the following are exclusively marine, **except**

(1) *Scolidon*

(2) *Trygon*

(3) *Torpedo*

(4) *Clarias*

194. Bony fishes and cartilaginous fishes are similar in possessing

(1) Air bladder

(2) Placoid scales

(3) Cycloid scales

(4) Two-chambered heart

195. In which of the following animals notochord persists throughout the life?

(1) *Carcharodon*

(2) *Ascidia*

(3) *Bufo*

(4) *Salpa*

196. Choose the correct match w.r.t. common name.

(1) *Chaetopleura* – Chiton

(2) *Pinctada* – Tusk shell

(3) *Wuchereria* – Hookworm

(4) *Pheretima* – Filaria worm

197. Match column I with column II on the basis of animals and their properties.

	Column I		Column II
a.	*Ctenoplana*	(i)	Proboscis
b.	*Branchiostoma*	(ii)	Bioluminescence
c.	*Saccoglossus*	(iii)	Lancelet
d.	*Myxine*	(iv)	Circular mouth

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(ii), b(iv), c(i), d(iii)

198. Metameric segmentation, open circulation with internal fertilisation is **not** found in

(1) *Anopheles*

(2) *Hirudinaria*

(3) *Periplaneta*

(4) *Ascaris*

199. Choose the **incorrect** match among the following:

(1)	*Balanoglossus*	–	Body is cylindrical and is composed of anterior proboscis, a collar and a long trunk
(2)	*Hippocampus*	–	Contains one auricle and one ventricle
(3)	*Naja*	–	Shows ecdysis
(4)	*Pheretima*	–	Excretes through Malpighian tubules

(1) (1)

(2) (2)

(3) (3)

(4) (4)

200. **Assertion :** Gaseous exchange in tadpole occurs through gills.
Reason : Pulmonary breathing has been evolved with terrestrial mode of habitat in animals.
In the light of above statements, select the correct option.

(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion and Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

Einstein3869